



Enhanced nitrogen prediction and mechanistic process analysis in high-salinity wastewater treatment using interpretable machine learning approach

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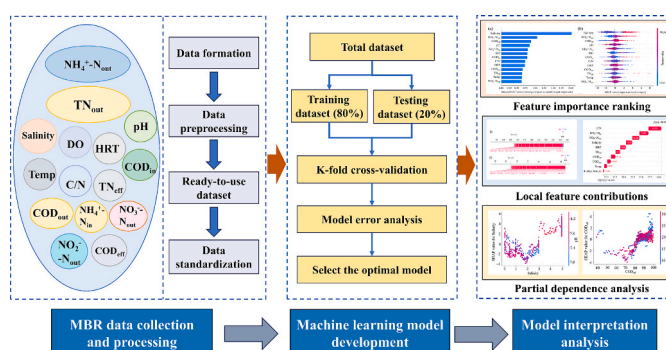
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HIGHLIGHTS

- CatBoost outperformed other models in nitrogen removal prediction.
- Salinity affects nitrifying enzyme reaction and carbon metabolism.
- Dissolved oxygen and COD efficiency were key nitrogen regulators.
- Temperature and C/N ratio affected total nitrogen via microbial activity.

GRAPHICAL ABSTRACT



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ABSTRACT

This study introduces an interpretable machine learning framework to predict nitrogen removal in membrane bioreactor (MBR) treating high-salinity wastewater. By integrating Shapley additive explanations (SHAP) with Categorical Boosting (CatBoost), we address the critical gap in linking predictive accuracy to operational decision-making for saline systems. CatBoost achieved the best performance, with an coefficient of determination (R^2) of 0.88 and root mean square error (RMSE) of 4.27 for the effluent ammonia nitrogen ($\text{NH}_4^+-\text{N}_{\text{out}}$), and an R^2 of 0.91 and RMSE of 4.35 for the effluent total nitrogen (TN_{out}). SHAP analysis uniquely revealed salinity's dual role in inhibiting nitrifying enzymes and disrupting carbon metabolism, with dissolved oxygen, pH and chemical oxygen demand removal efficiency as key regulators. Temperature and carbon-to-nitrogen ratio further modulated total nitrogen dynamics through electron donor availability and microbial activity. The proposed SHAP-CatBoost model in high salinity MBR combines predictive modelling with mechanical process control.

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1. Introduction

With the rapid advancement of urbanization and industrialization, water scarcity and pollution issues have intensified (Bao and Fang, 2012). Chemical manufacturing, food processing, pharmaceuticals, and seawater desalination commonly produce high-salinity wastewater (Guo et al., 2023). Elevated salt concentrations pose severe environmental threats, including soil permeability degradation and aquatic life toxicity (Linarić et al., 2013). Membrane bioreactors (MBRs) combine membrane separation with biological treatment, making them an ideal technology for treating high-salinity wastewater (Neoh et al., 2016).

In MBR operations, wastewater treatment plants (WWTPs) face significant salinity-driven challenges. High NaCl (>1%) suppresses ammonia-oxidizing bacteria (AOB) activity, necessitating costly chemical supplementation to maintain nitrogen compliance (Zhang and Liu, 2023). Salinity-stressed microbes prioritize osmoprotectant synthesis over denitrification, but existing models fail to quantify these carbon flux trade-offs (Jeon et al., 2022). Additionally, traditional sensors cannot detect salinity-induced microbial shifts in real time, leading to delayed aeration adjustments and excessive energy consumption (Waki et al., 2018). These issues inflate operational costs by 18–25 % in WWTPs, highlighting the need for interpretable models linking predictions to actionable process controls (Borzooei et al., 2019).

Traditional MBR approaches like mechanistic models, empirical regressions, and hybrid frameworks have succeeded in low-salinity wastewater scenarios. For example, Park et al. (2015) developed a transmembrane pressure-driven flux model predicting fouling rates, while Waki et al. (2018) optimized aeration cycles using activated sludge models (ASM), reducing energy consumption by 18 %. However, these models struggle with high-salinity wastewater. Mechanistic models suffer from kinetic parameter drift when applied to saline conditions (Huang et al., 2020), and hybrid models like ASM coupled with computational fluid dynamics (ASM-CFD) coupling overpredict nitrification rates at elevated salinity (Xu et al., 2024). Empirical models fail to generalize across varying salinity gradients, as osmotic stress disrupts carbon-to-nitrogen (C/N) metabolic pathways (Zhang et al., 2015).

Recent advances in machine learning (ML) have enhanced predictive capabilities in wastewater treatment. Neural networks, such as artificial neural networks (ANN), have been used to predict total nitrogen (TN) removal (Jeon et al., 2022), while long short-term memory (LSTM) models handle dynamic water quality data (Yaqub et al., 2020). Tree-based models like XGBoost optimize chemical oxygen demand (COD) removal (Yaqub and Lee, 2022), and random forests (RF) identify heavy metal toxicity thresholds (Gaur et al., 2024). Hybrid models, such as ASM-ANN coupling, demonstrate the potential of machine learning in wastewater treatment (Xu et al., 2024).

Research on high-salinity and heavy metal wastewater has predominantly focused on membrane fouling, with limited attention on microbial metabolic processes (Borzooei et al., 2019). Studies on heavy metal wastewater have emphasized feature selection (e.g., pH, ion strength) yet often overlook process control integration (Arévalo et al., 2013). Notably, no study has employed SHAP's force plots or partial dependence plots (PDPs) to explore the salinity-induced trade-offs between nitrification enzyme activities and carbon allocation in MBRs. This gap limits the practical application of ML models in hypersaline environments, where threshold-driven control strategies are crucial for optimal performance.

This study bridges these gaps by developing an interpretable ML framework that quantifies and prioritizes salinity-impacted control variables in MBRs. The objectives are: 1) to establish an interpretable machine learning framework that decodes salinity-driven nitrogen removal bottlenecks; 2) to quantify multiscale interactions between operational parameters and microbial nitrification-denitrification functions under high-salt stress; and 3) to develop decision-centric visual tools that translate model predictions into actionable control strategies for WWTPs. This approach advances beyond conventional ML

applications by embedding biological mechanism transparency, offering a paradigm shift toward intelligent, cost-effective MBR optimization.

2. Materials and methods

2.1. Membrane reactor equipment

This study aims to simulate real-world conditions in MBRs treating high-salinity wastewater, focusing on salinity-induced metabolic stress and its impact on nitrogen removal efficiency. The experimental conditions were selected to reflect typical operational challenges in high-salt environments. The synthetic wastewater was prepared using glucose ($C_6H_{12}O_6$), potassium dihydrogen phosphate (KH_2PO_4), ammonium chloride (NH_4Cl), sodium chloride ($NaCl$), and trace elements based on previous studies (Khan et al., 2011; Zhong et al., 2022). Five batches of reactors, each with an adequate volume of 10 L, were cultivated at salinity levels ranging from 0 to 5 ‰. Regulate the temperature between 5 and 35 °C using a water bath heating system, maintaining suspended solids (SS) concentrations between 3 and 8 g/L and pH values ranging from 7.0 to 8.5. The peristaltic pump achieves continuous water inlet and outlet, the hydraulic retention time (HRT) is between 19 and 30 h, and the dissolved oxygen (DO) level is between 0.1 and 3.5 mg/L. The influent total nitrogen (TN_{in}) consisted of ammonia nitrogen ($NH_4^+-N_{in}$), ranging from 20 to 90 mg/L. While the initial carbon source concentration of chemical oxygen demand (COD) equivalent varied between 180 and 300 mg/L. Hollow fibre membrane components were employed, and membrane fouling was monitored using transmembrane pressure gauges. The treatment procedure in each reactor module, whereby wastewater was introduced into the MBR using a peristaltic pump, agitated by a stirring and aeration system. Organic matter was biodegraded by denitrification and activated sludge process. Discharge the treated effluent via the membrane modules (Fig. 1).

2.2. Data collection and processing

Evaluate the quality of the treated water following standard procedures for measuring various nitrogen compounds and organic content. Monitor key parameters such as DO and pH using a portable meter designed for field use (HI98191, Hanna®, Italy). Adjustments to HRT were made by controlling the peristaltic pumping device's flow rate. Systematically organized and analyzed the collected monitoring data resulted in 15 key feature parameters, including salinity, oxygen levels, retention time, acidity, temperature, and input and output concentrations of key contaminants. Specifically, the output two parameters were the effluent of ammonia nitrogen ($NH_4^+-N_{out}$) and total nitrogen (TN_{out}), while the remaining 13 parameters served as input variables: salinity, DO, HRT, pH, temperature (Temp), chemical oxygen demand in the influent (COD_{in}), chemical oxygen demand in the effluent (COD_{out}), ammonia nitrogen in the influent ($NH_4^+-N_{in}$), nitrite nitrogen in the effluent ($NO_2^- - N_{out}$), nitrate nitrogen in the effluent ($NO_3^- - N_{out}$), carbon-to-nitrogen (C/N) ratio, chemical oxygen demand removal efficiency (COD_{eff}) and total nitrogen removal efficiency (TN_{eff}) in the effluent of MBR.

The pre-processing steps include removing duplicate samples and refining the data set to ensure the accuracy of the model and prevent data leakage. Input features were optimized using standardization techniques provided by the Scikit-learn library. The Z-score method detects outliers and removes extreme values above the threshold of 3. Additionally, winsorization was applied to cap values outside the 1st and 99th percentiles, preserving the data's overall distribution. The dataset was then randomly split, with 80 % allocated for training and 20 % reserved for testing. The final processed dataset consisted of 570 samples (see supplementary materials). Fig. 2a shows a box plot of all the data, giving the mean and standard deviation. All machine learning models were validated using this testing set, allowing for a comprehensive accuracy evaluation. 10-fold cross-validation was applied in the

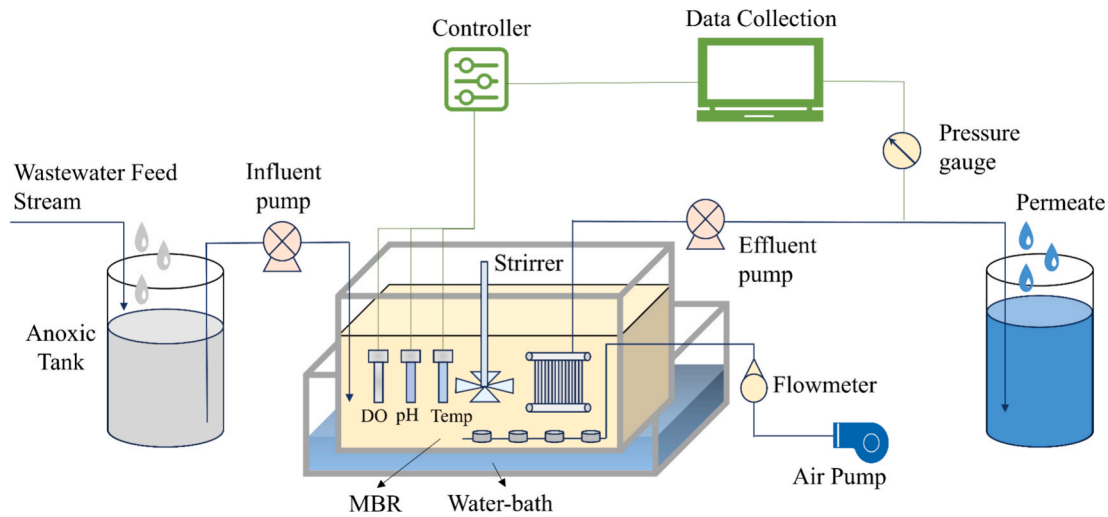


Fig. 1. Schematic diagram of membrane bioreactor experimental apparatus.

training phase to ensure the robustness of the model evaluation. This method divides the data into 10 subsets, iterating through 10 rounds. Train with nine subsets, validate with the remaining subsets, and train and test with each data point. This approach helps minimize overfitting and provides a more reliable estimate of model performance. Correlation coefficients ranged from -0.05 to 0.22 for $\text{NH}_4^+-\text{N}_{\text{out}}$ and from -0.52 to 0.26 for TN_{out} (Fig. 2b). Pearson correlational analysis revealed weak associations between nitrogen-related outputs and other variables, indicating their lower influence on the overall model performance.

2.3. Model construction and optimization

2.3.1. Machine learning algorithms

This study applies six machine learning models to simulate effluent water quality. Gradient boosting decision trees (GBDT) is an ensemble learning method that sequentially builds multiple weak decision trees, with each tree correcting the errors of the previous one (Niu et al., 2023). Light gradient boosting machine (LightGBM) is a variant of GBDT, using a histogram-based approach to speed up training and improve efficiency (Chang et al., 2022). Random forest (RF) is an integrated approach to building multiple decision trees, each trained on a random subset of data and features (Lai et al., 2025). XGBoost is another gradient-boosting algorithm that optimizes performance and computational efficiency using techniques like regularization and parallelization (Yaqub and Lee, 2022). Adaptive boosting (AdaBoost) is an ensemble method that combines weak classifiers in a weighted manner to form a strong classifier, giving more weight to incorrectly classified instances in subsequent rounds (Zhuang et al., 2021). Categorical boosting (CatBoost) is a gradient-boosting algorithm that efficiently handles categorical features by converting them into numerical representations during training (Ibrar et al., 2022). These models were applied using 13 experimental input variables to predict effluent $\text{NH}_4^+-\text{N}_{\text{out}}$ and TN_{out} in the MBR.

2.3.2. Bayesian optimization

Fine-tuning the hyperparameters of each model using Bayesian optimization improves performance while reducing computational costs. It builds a probabilistic model, typically a Gaussian process, to predict the performance of different hyperparameter configurations, guiding the selection of optimal values through a balance of exploration and exploitation (Shi et al., 2021; Yin and Li, 2022). The search space for hyperparameters, such as learning rate, tree depth, and iteration number, was defined based on prior knowledge and preliminary results. Compared to grid and random search, Bayesian optimization explores

the parameter space more efficiently, achieving better results with fewer evaluations. Comparative experiments with stochastic and grid search confirmed that Bayesian optimization outperformed these traditional methods in terms of both accuracy and computational efficiency, making it the most suitable choice for our specific application.

In the context of Bayesian inference, we aim to estimate the posterior distribution of a function f given observed data D . The observed data D is a set of input–output pairs, and the relationship between the input and output can be modeled as:

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \quad (1)$$

$$y_i = f(x_i) + \epsilon_i \quad (2)$$

where x_i denotes the decision vectors, y_i represents the corresponding outputs, and ϵ_i represents the observation error.

By applying Bayes' theorem, the posterior probability distribution $p(f|D)$ of the unknown target function $f(x)$ can be expressed as:

$$p(f|D) = \frac{p(D|f) \cdot p(f)}{p(D)} \quad (3)$$

where $p(D|f)$ is the likelihood distribution of observed data D , $p(f)$ is the prior probability distribution of $f(x)$, $p(D)$ is the marginal likelihood of $f(x)$ to optimize hyperparameters.

2.3.3. K-fold cross-validation

For model validation, K-fold cross-validation was employed to reduce bias and ensure stability across the datasets. This approach splits the data into k ($k = 10$) equally sized portions. Designate one portion as the test set for each iteration while using the remaining portions for training. This procedure is repeated k times, ensuring each subset is precisely set once as the test (Fushiki, 2011; Zhang and Liu, 2023). The average performance across all folds provides a robust measure of the model's generalization ability, reducing overfitting and variance in our results.

2.3.4. Implementation of machine learning models

The models were implemented using Python 3.11.4 and scikit-learn libraries. The datasets were preprocessed to ensure compatibility with each algorithm, including scaling and encoding as necessary. Each model was trained using optimized hyperparameters obtained from Bayesian optimization, and cross-validated results were used to ensure consistency in model performance evaluation.

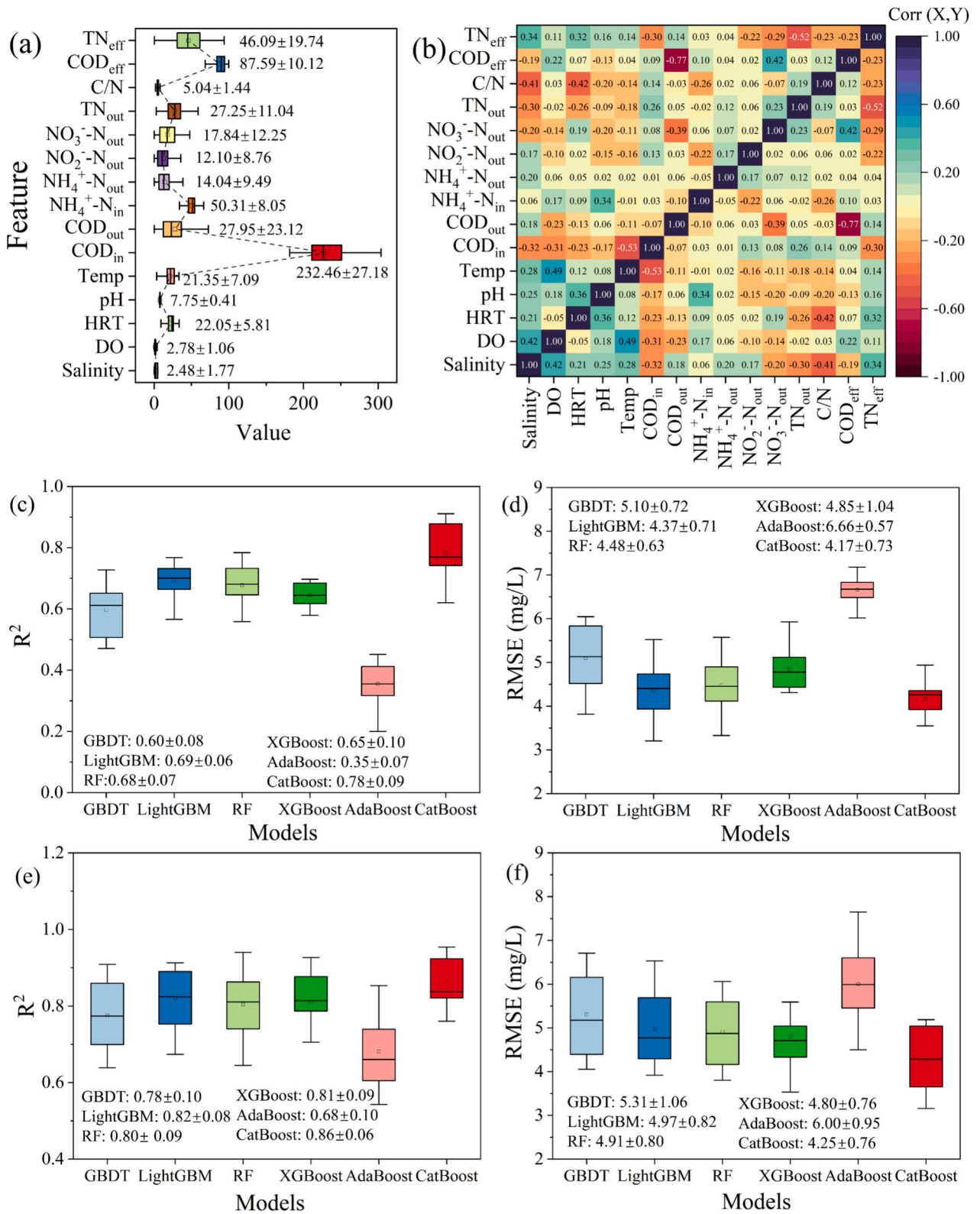


Fig. 2. (a) Range of values for each feature; (b) Heatmaps for different parameters; 10-fold cross-validation results for six machine learning algorithms: (c) R² and (d) RMSE for the NH₄⁺-N_{out} modeling; (e) R² and (f) RMSE for the TN_{out} modeling.

2.4. Model performance evaluation

The performance of each machine learning model was evaluated using coefficient of determination (R²) and root mean square error (RMSE). The equations are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\bar{y} - y_i)^2} \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (5)$$

where n represents the sample amount; y_i and $\hat{y}_i$ are the simulated and measured values; $\bar{y}$ is the average of the measured values. They were computed for all models using the test dataset after applying cross-validation to ensure robust and unbiased results.

2.5. Model interpretation

Shapley additive explanations (SHAP) analysis was employed to enhance the interpretability of all machine learning models. SHAP values offer a game-theoretic method for explaining model outputs, highlighting the contribution of each feature to the predictions. These values are derived from Shapley values, where the contribution of each feature is determined by considering all possible feature combinations. The SHAP value for each factor i is calculated using the expression:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i \quad (6)$$

where $g(z')$ is the explanation model; M is the total number of features; ϕ_i is the Shapley value of feature i ; $z'_i \in \{0, 1\}$ represents whether a feature is included or excluded in the Shapley value calculation.

SHAP summary plot displays the overall importance of features across all predictions. It ranks features based on their contribution to the model's output, providing a global view of the most influential parameters. Force plot visual explains individual predictions by showing the force each feature exerts on the prediction, pushing it higher or lower than the average prediction (He et al., 2023). The waterfall plot visualizes how each feature contributes to a specific prediction by showing a cumulative breakdown of SHAP values. It allows us to understand the direction and magnitude of feature influence on a particular instance. The PDPs help illustrate the relationship between one or two features and the target variable. It shows how the expected prediction changes as a feature varies, holding other features constant, and provides deeper insights into feature behaviour (Kovacs et al., 2022; Niu et al., 2024). By combining these visual tools, global and local insights were gained into the machine learning models, improving the overall interpretability of the predictions.

3. Results and discussions

3.1. Model training and optimization

3.1.1. Hyperparameter optimization

Bayesian optimization with Gaussian processes was applied to tune hyperparameters across six models, utilizing a search space defined by prior experimental knowledge and pilot trials. For each algorithm, key parameters were bounded within physiochemical and computational constraints: learning rates (0.01–0.30) to balance convergence speed and stability, tree depths (3–20) to prevent overfitting, and regularization terms (L2: 1–10) to control model complexity. The optimization process iteratively evaluated different parameter combinations per model via a lower confidence bound (LCB) acquisition function, maximizing the expected improvement in 10-fold cross-validation RMSE. CatBoost's depth (4–10) and L2 regularization (1–10) were optimized to mitigate overfitting under high-salinity noise, converging at depth = 8 and L2 = 3. LightGBM's leaf count (20–50) and RF's tree depth (5–20) addressed feature interaction capture in nonlinear salinity-response systems. XGBoost's subsampling (0.50–1.00) enhanced robustness against ionic strength-induced data sparsity (see [supplementary materials](#)). The algorithm-specific ranges ensured physiochemical relevance with known microbial metabolic hierarchies. Optimal configurations reduced validation RMSE by 12–18 % compared to default settings,

confirming Bayesian optimization's efficacy in hypersaline wastewater prediction tasks.

3.1.2. Model performance comparison

Evaluate the results of the 10-fold cross-validation for the six machine learning algorithms applied to model $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} concentrations. The performance metrics, including R^2 and RMSE, for $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} modelling were summarized in Table 1. CatBoost outperformed all other models, achieving an R^2 of 0.88 and an RMSE of 4.27 for $\text{NH}_4^+\text{-N}_{\text{out}}$, as well as an R^2 of 0.91 and an RMSE of 4.35 for TN_{out} . LightGBM, the second-best model, delivered an R^2 of 0.81 and an RMSE of 4.97 for $\text{NH}_4^+\text{-N}_{\text{out}}$, and an R^2 of 0.82 and an RMSE of 5.17 for TN_{out} . The other models, including GBDT, RF, XGBoost, and AdaBoost, demonstrated lower performance metrics. Compared with LightGBM, the CatBoost improved the model's performance, increasing R^2 by 9 % for $\text{NH}_4^+\text{-N}_{\text{out}}$ and 10 % for TN_{out} , while reducing RMSE by 14 % and 16 %, respectively.

Fig. 2c and 2d illustrate the R^2 and RMSE results for $\text{NH}_4^+\text{-N}_{\text{out}}$ modelling, while Fig. 2e and 2f display the corresponding results for TN_{out} modelling. These visualizations further emphasize CatBoost's superior performance in predicting both nitrogen species. These plots reveal that CatBoost's predictions closely align with actual measurements, indicating its effectiveness in modelling nitrogen concentrations. Overall, the results justify the selection of CatBoost as the optimal model for predicting effluent nitrogen.

3.2. Feature importance ranking

The SHAP summary plot elucidates the relative contributions of input variables to the predictions of $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} , revealing distinct mechanistic drivers for each output parameter (Fig. 3). The selection of the top six predictors for $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} was based on the synergy between SHAP rankings and mechanistic interpretability, prioritizing parameters with both high model influence and actionable process relevance.

For $\text{NH}_4^+\text{-N}_{\text{out}}$ predictions, salinity is identified as the most influential factor, with a SHAP value nearly twice that of the second-ranked variable, $\text{NO}_2^-\text{-N}_{\text{out}}$, indicating its substantial impact. The other key predictors include $\text{NO}_2^-\text{-N}_{\text{out}}$, COD_{eff} , pH, $\text{NH}_4^+\text{-N}_{\text{in}}$, and DO. While DO, COD_{in} , and C/N ratio have similar SHAP values, DO was prioritized due to its direct role in regulating nitrification and denitrification, making it more practical for process control (Fig. 3a and Fig. 3b). Salinity exerts primary control over ammonium accumulation by directly suppressing nitrifying bacteria activity. Elevated salt concentrations disrupt transmembrane proton gradients in AOB, inhibiting ammonia monooxygenase (AMO) and hydroxylamine oxidoreductase (HAO) enzymes critical for NH_4^+ oxidation (Yan et al., 2018). $\text{NO}_2^-\text{-N}_{\text{out}}$ accumulation exacerbates nitrification stagnation through feedback inhibition, as nitrite binds to AMO active sites competitively. COD_{eff} further modulates $\text{NH}_4^+\text{-N}_{\text{out}}$ by reflecting carbon source competition: high COD removal efficiency implies heterotrophic dominance, depriving nitrifiers of essential carbon substrates for energy metabolism (Bai et al., 2023). pH fluctuations (optimal range: 7.5–8.5) destabilize proton pump equilibrium in nitrifying consortia, particularly under salinity-induced ionic

Table 1
Error analysis of different machine learning models.

$\text{NH}_4^+\text{-N}_{\text{out}}$			TN_{out}		
Models	R^2	RMSE (mg/L)	Models	R^2	RMSE (mg/L)
CatBoost	0.88	4.27	CatBoost	0.91	4.35
LightGBM	0.81	4.97	LightGBM	0.82	5.17
RF	0.79	5.48	RF	0.80	5.25
XGBoost	0.71	5.85	XGBoost	0.76	5.20
GBDT	0.68	6.10	GBDT	0.71	5.71
AdaBoost	0.54	7.66	AdaBoost	0.63	6.50

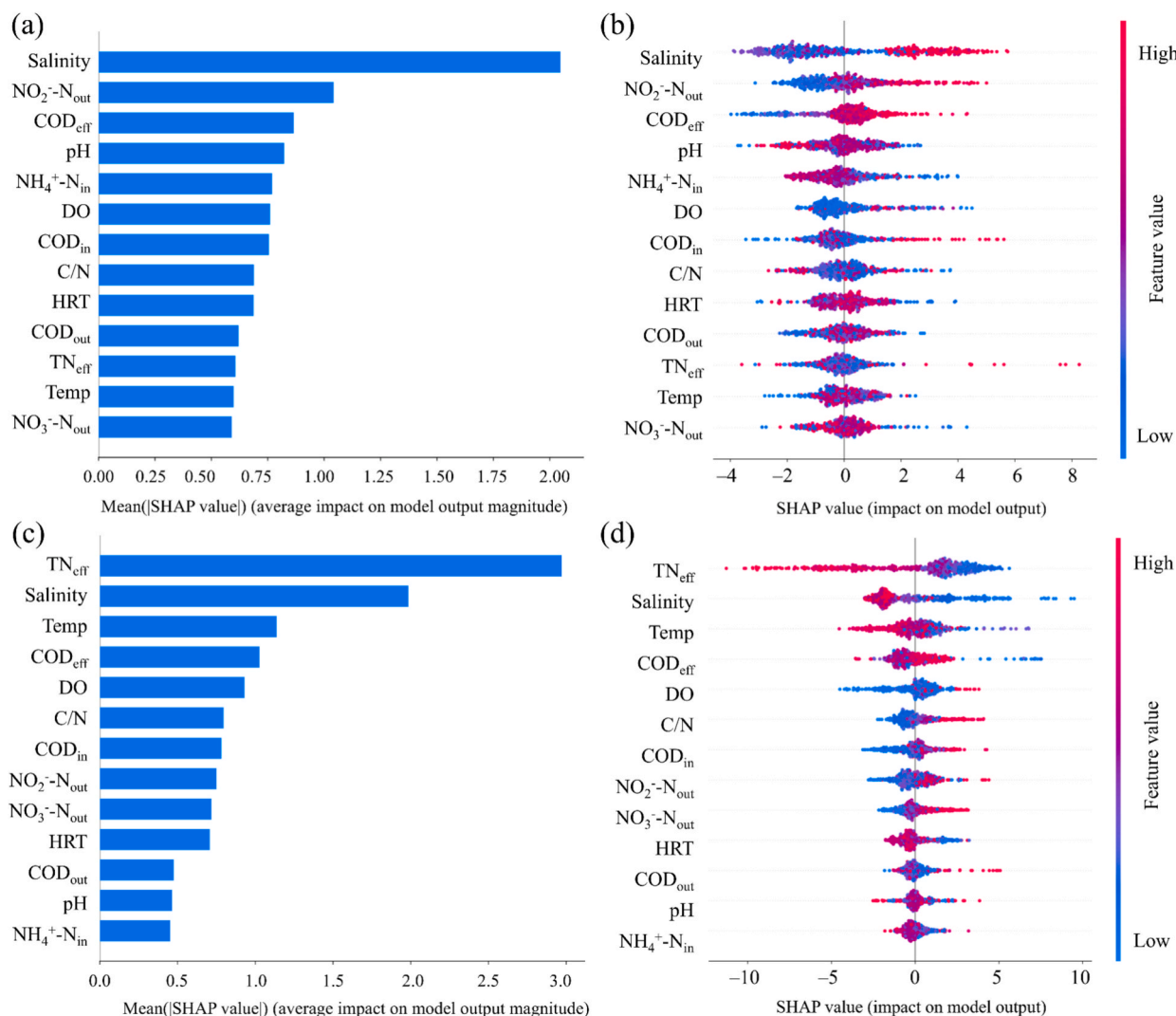


Fig. 3. Interpretable analysis of $\text{NH}_4^+\text{-N}_{\text{out}}$ (a-b) and TN_{out} (c-d) for CatBoost model.

strength variations. $\text{NH}_4^+\text{-N}_{\text{in}}$ directly governs reactor ammonium loading; concentrations exceeding nitrification capacity lead to proportional increases in effluent $\text{NH}_4^+\text{-N}_{\text{out}}$. DO demonstrate nonlinear effects—nitrification requires > 2 mg/L dissolved oxygen, but salinity-driven reductions in oxygen solubility or accelerated microbial respiration may create localized hypoxia, constraining ammonia oxidation (Huang et al., 2023). In contrast, TN_{eff} , temperature, and $\text{NO}_3^-\text{-N}_{\text{out}}$ exhibit minimal influence on $\text{NH}_4^+\text{-N}_{\text{out}}$ predictions. TN_{eff} predominantly reflects denitrification efficiency, which decoupled from nitrification-dominated NH_4^+ accumulation. Temperature impacts are marginal within the tested range (15–30 °C), likely masked by salinity stress. $\text{NO}_3^-\text{-N}_{\text{out}}$ shows weak correlations due to nitrification arrest at the nitrite phase under high salinity (Luo et al., 2019).

For TN_{out} , TN_{eff} has the highest SHAP contribution, as expected due to its strong correlation with TN_{out} , followed by salinity, which has a SHAP value about two-thirds of TN_{eff} . The remaining four dominant predictors include temperature, COD_{eff} , DO, and the C/N ratio. Although COD_{in} and $\text{NO}_2^-\text{-N}_{\text{out}}$ have SHAP values close to the C/N ratio, the selection of these six predictors was not solely based on SHAP rankings but also on their practical implications in MBR operation. Specifically, DO and the C/N ratio were prioritized over COD_{in} and $\text{NO}_2^-\text{-N}_{\text{out}}$ due to their critical roles in regulating microbial activity and nitrogen transformation pathways (Fig. 3c and Fig. 3d). TN_{eff} emerges as the strongest predictor, directly quantifying systemic nitrogen removal performance.

Salinity exerts dual inhibition: it disrupts both nitrification (limiting $\text{NH}_4^+\text{-N}_{\text{in}}$ conversion) and denitrification by suppressing nitrate reductase (Nar) and nitrite reductase (Nir) activities (Qiang et al., 2023). Salt-stressed microbes prioritize compatible solute synthesis (e.g., glycine betaine) over metabolic functions, diverting carbon resources from denitrification electron donors. Temperature sensitivity in TN_{out} predictions aligns with the thermophilic nature of denitrifiers (optimal range: 20–40 °C). Low temperatures (< 15 °C) inactivate Nar enzymes, stalling NO_3^- reduction. COD_{eff} and C/N ratio jointly regulate carbon availability—high COD removal efficiency or low C/N ratios (< 5) starve denitrifiers of organic electron donors, leaving residual $\text{NO}_3^-/\text{NO}_2^-$ unresolved (Zhang et al., 2015). DO exhibits threshold-dependent effects: elevated DO (> 0.5 mg/L) in anoxic zones inhibits Nar activity, with salinity-induced aeration heterogeneity exacerbating oxygen intrusion. Notably, COD_{out} , pH, and $\text{NH}_4^+\text{-N}_{\text{in}}$ show limited relevance. COD_{out} primarily reflects terminal COD degradation, offering no incremental insight beyond COD_{eff} . Denitrifiers tolerate broader pH ranges (6.5–8.5), and salinity's ionic effects may buffer pH fluctuations. $\text{NH}_4^+\text{-N}_{\text{in}}$'s diminished role for TN_{out} depends on post-nitrification intermediates ($\text{NO}_3^-/\text{NO}_2^-$) and their subsequent denitrification (Li et al., 2022).

3.3. Local feature contributions

The SHAP force plot visually demonstrated how individual input

factors impact the model's prediction, showing the cumulative effect leading to the final prediction outcome. For $\text{NH}_4^+\text{-N}_{\text{out}}$ prediction, C/N ratio, $\text{NH}_4^+\text{-N}_{\text{in}}$, $\text{NO}_3^-\text{-N}_{\text{out}}$, salinity, HRT and TN_{eff} are among the most influential parameters, with COD_{eff} , COD_{out} and temperature showing substantial contributions. Each feature either pushes the prediction higher, illustrating their role in the nitrification process (Fig. 4a). For TN_{out} prediction, similar patterns were observed. The force plot highlights salinity, TN_{eff} , temperature, COD_{eff} , $\text{NO}_2^-\text{-N}_{\text{out}}$, and C/N ratio as key factors contributing to the model's prediction. DO and pH slightly increase the predicted value of TN, while salinity and TN_{eff} have a more pronounced influence. The relatively significant contributions of the C/N ratio and $\text{NO}_2^-\text{-N}_{\text{out}}$ further emphasize the sensitivity of TN removal to these factors (Fig. 4b).

The SHAP waterfall plot provides a more granular breakdown of the individual feature contributions to a specific prediction. For $\text{NH}_4^+\text{-N}_{\text{out}}$, the key parameters contributing to the final prediction include C/N ratio, $\text{NH}_4^+\text{-N}_{\text{in}}$, $\text{NO}_3^-\text{-N}_{\text{out}}$, and salinity. These features cumulatively increase the predicted $\text{NH}_4^+\text{-N}_{\text{out}}$, suggesting that higher influent ammonia concentrations, nitrate levels, and salinity directly correlate with higher effluent ammonia levels. Operational parameters such as HRT, TN_{eff} and COD_{eff} also contribute positively but to a lesser extent, indicating their role in nitrogen removal efficiency under the given conditions (Fig. 4c). For TN_{out} , the waterfall plot shows that salinity, TN_{eff} , and temperature are the dominant factors increasing the predicted value. COD_{eff} , $\text{NO}_2^-\text{-N}_{\text{out}}$, and C/N ratio also contribute positively, though to a lesser degree. The impact of salinity on TN_{out} highlights the inhibitory effects on nitrogen removal processes, where high salinity can limit denitrifying microbial activity (Fig. 4d).

3.4. Partial dependence analysis

Employ PDPs to visualize how changes in individual features influence the predictions of $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} . PDPs allow us to explore the direction and magnitude of the impact of key factors like salinity, COD_{eff} , and pH on the predicted values of $\text{NH}_4^+\text{-N}_{\text{out}}$ and TN_{out} . The non-linear relationships between features and output variables are revealed through these plots, providing deeper insights into parameter behaviour. Colour coding represents the interaction effects of other variables, with red representing higher feature values and blue indicating lower values. This method helps to visually distinguish the effects of different feature levels on model prediction.

For $\text{NH}_4^+\text{-N}_{\text{out}}$, we focus on the top six features identified in the summary plot. The partial dependence of salinity on $\text{NH}_4^+\text{-N}_{\text{out}}$ exhibits a downward trend followed by a sharp increase. When salinity is below 2 ‰, the $\text{NH}_4^+\text{-N}_{\text{out}}$ decreases with increasing salinity, and $\text{NH}_4^+\text{-N}_{\text{out}}$ gradually increases when salinity is above 2 ‰ (Fig. 5a). This suggests that low salinity levels are conducive to better ammonia removal. However, as salinity increases beyond this point, the SHAP values of $\text{NH}_4^+\text{-N}_{\text{out}}$ increase due to the inhibitory effect of high salinity on microbial nitrification processes. The effect of $\text{NO}_2^-\text{-N}_{\text{out}}$ on the model's predictions shows a consistent upward trend (Fig. 5b), suggesting that higher nitrite concentrations lead to increased $\text{NH}_4^+\text{-N}_{\text{out}}$. This phenomenon indicates that nitrite accumulation in sewage indicates that denitrification is incomplete or that the nitrification process is interrupted. The influence of COD_{eff} on $\text{NH}_4^+\text{-N}_{\text{out}}$ is minimal when COD_{eff} is below 70 %. However, once COD_{eff} exceeds this threshold, the concentration of $\text{NH}_4^+\text{-N}_{\text{out}}$ rises significantly (Fig. 5c). This relationship emphasizes the importance of maintaining lower organic load levels to optimize ammonia removal efficiency.

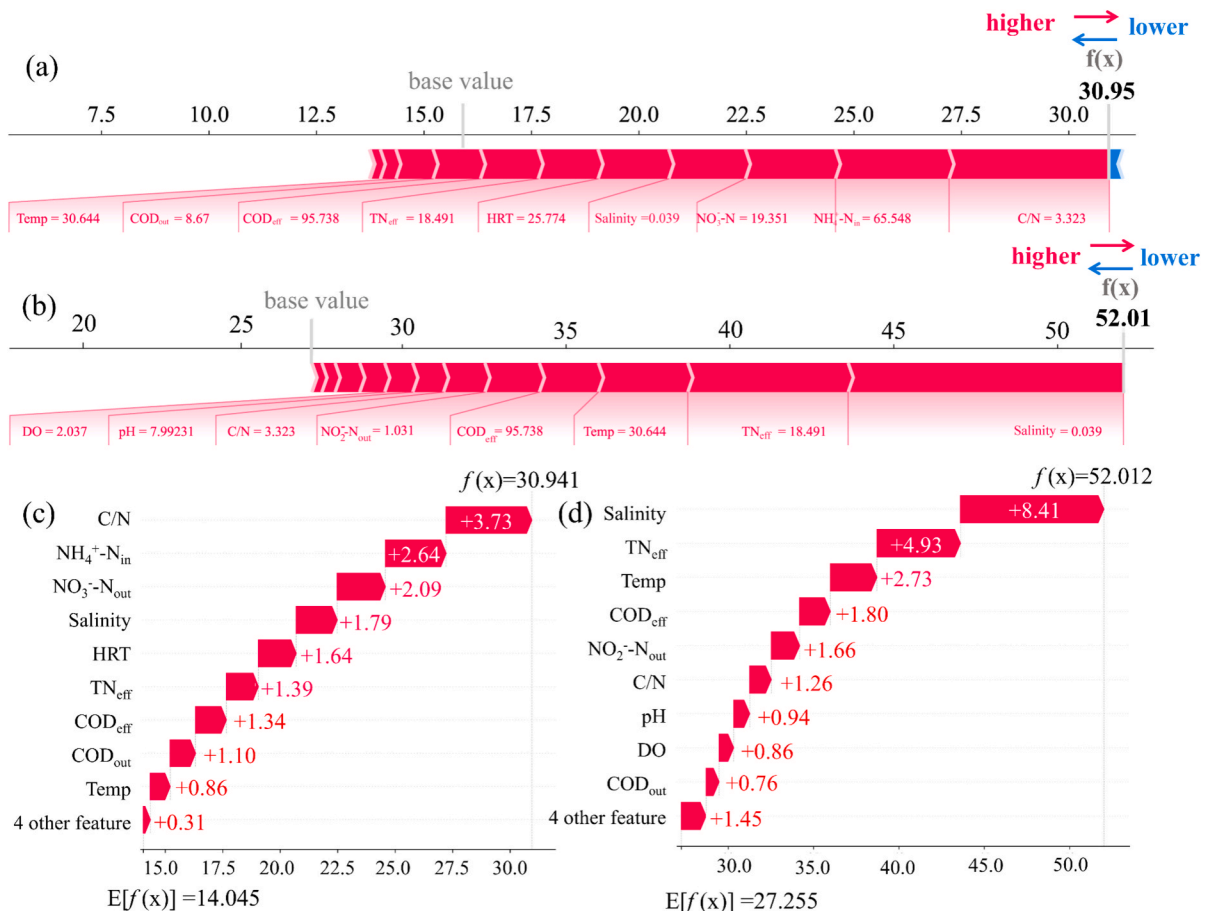


Fig. 4. Interpretation analysis using CatBoost and SHAP: (a) $\text{NH}_4^+\text{-N}_{\text{out}}$ force plot; (b) TN_{out} force plot; (c) $\text{NH}_4^+\text{-N}_{\text{out}}$ waterfall plot; (d) TN_{out} waterfall plot.

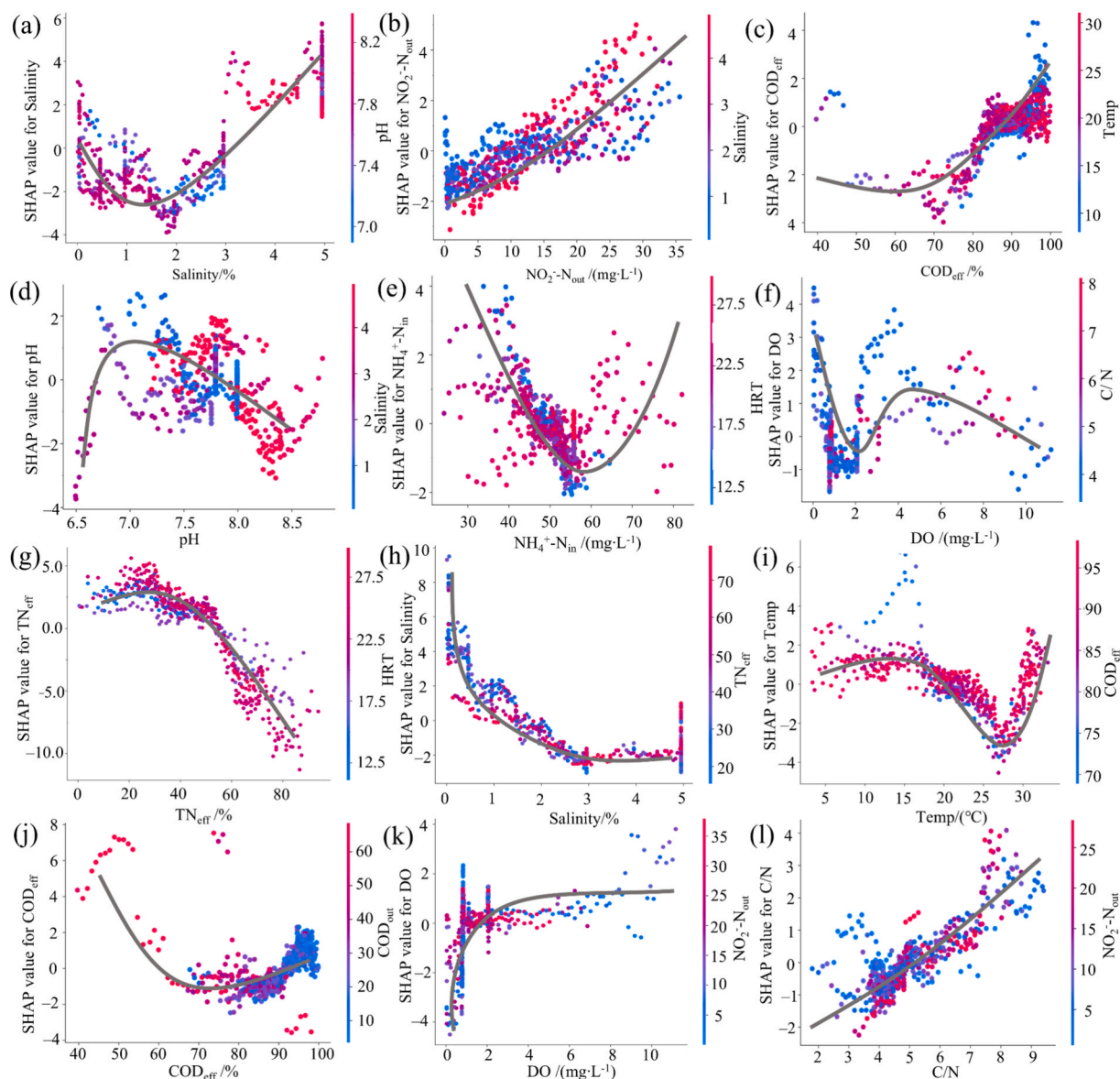


Fig. 5. The partial dependence plots of first six feature based on CatBoost model for $\text{NH}_4^+\text{-N}_{\text{out}}$ (a-f) and TN_{out} (g-l). Note: (a) salinity; (b) $\text{NO}_2^-\text{-N}_{\text{out}}$; (c) COD_{eff} ; (d) pH; (e) $\text{NO}_2^-\text{-N}_{\text{out}}$; (f) DO; (g) TN_{eff} ; (h) salinity; (i) temperature; (j) COD_{eff} ; (k) DO; (l) C/N.

The pH's impact on $\text{NH}_4^+\text{-N}_{\text{out}}$ increases rapidly as the pH approaches 7.0 and begins to decrease once the pH exceeds 7.5 (Fig. 5d). The SHAP value becomes negative beyond this point, reducing the model's predicted probability of $\text{NH}_4^+\text{-N}_{\text{out}}$ occurrence. The interaction effect with salinity becomes most pronounced at pH values greater than 8.0. When the $\text{NH}_4^+\text{-N}_{\text{in}}$ is below 55 mg/L, SHAP values remain positive, suggesting a direct relationship with $\text{NH}_4^+\text{-N}_{\text{out}}$ reduction. However, when $\text{NH}_4^+\text{-N}_{\text{in}}$ exceeds 60 mg/L, the concentration of $\text{NH}_4^+\text{-N}_{\text{out}}$ begins to rise again (Fig. 5e). This indicates that excessively high influent ammonia concentrations can hinder the removal efficiency. When DO concentrations are between 0–2 mg/L, SHAP values decrease rapidly, meaning low DO levels reduce the $\text{NH}_4^+\text{-N}_{\text{out}}$. However, the SHAP value rose when DO increased to 2–6 mg/L. When DO exceeded 6 mg/L, the SHAP value gradually decreased, indicating that insufficient DO may lead to inadequate nitrogen removal reaction, and higher DO is more conducive to ammonia nitrogen removal (Fig. 5f).

The PDPs provide insights into the influence of key parameters on the prediction of TN_{out} . The analysis reveals distinct patterns for six critical features. TN_{eff} generally correlates negatively with TN_{out} concentration, particularly when TN_{eff} exceeds 40 %. As the nitrogen

removal efficiency increases, the total nitrogen concentration significantly decreases (Fig. 5g). Furthermore, the interaction between TN_{eff} and HRT becomes most pronounced when TN_{eff} ranges from 20 % to 60 %. A clear non-linear negative correlation is observed between salinity and TN_{out} . Specifically, when salinity is between 0 and 2 %, an increase in salinity leads to a notable reduction in TN_{out} (Fig. 5h). However, when salinity surpasses 2 %, the SHAP values stabilize, indicating a diminishing effect on nitrogen removal beyond a certain threshold. This behaviour aligns with the sensitivity of nitrifying and denitrifying bacteria to changes in salinity. The influence of temperature on TN_{out} removal is minimal when temperatures range from 0 °C to 20 °C, as the SHAP values remain relatively stable. However, as the temperature rises from 20 °C to 25 °C, the SHAP values gradually decrease, and a sharp increase is seen beyond 30 °C. This phenomenon suggests that a temperature range of 25–30 °C is optimal for the metabolic activity of nitrifying bacteria (Fig. 5i). Additionally, the interaction between temperature and COD_{eff} is particularly significant within the 15–30 °C range, indicating that the organic load influences the impact of temperature on TN_{out} removal.

When COD_{eff} is below 70 %, a linear negative correlation is observed

between COD_{eff} and TN_{out} , indicating that as the COD removal efficiency increases, the TN_{out} concentration gradually decreases. However, when COD_{eff} exceeds 70 %, the SHAP values begin to increase, suggesting that a reduction in the carbon source may diminish the metabolic activity of nitrifying and denitrifying bacteria (Fig. 5j). Regarding DO, when it is between 0 and 2 mg/L, a significant increase in SHAP values is observed as DO rises, indicating that higher DO concentrations enhance the TN_{out} levels. However, beyond 2 mg/L, the SHAP values tend to stabilize, implying that further increases in DO have diminishing effects. The interaction between NO_2^- - N_{out} and DO is most pronounced when DO is between 0 and 4 mg/L (Fig. 5k). Elevated levels of DO are detrimental to the anaerobic ammonia oxidation process, suggesting that maintaining a lower DO concentration is more favourable for TN_{out} removal. A positive correlation between the C/N ratio and TN_{out} is observed, where higher C/N ratios inhibit denitrification, indicating that the ratio should be kept at a lower level for optimal nitrogen removal (Fig. 5l). The interaction between the C/N ratio and NO_2^- - N_{out} is most significant when the C/N ratio is between 3 and 8 mg/L, suggesting that this range is most conducive to nitrification activity.

3.5. Underlying mechanisms and processes

In high-salinity environments, NH_4^+ - N_{out} removal is primarily hindered by the inhibition of nitrification. The activity of ammonia-oxidizing bacteria (AOB) and nitrite-oxidizing bacteria (NOB) is sensitive to changes in osmotic pressure caused by salt (Shen et al., 2014). This osmotic stress disrupts cellular functions, including the proton gradient required for the activity of AMO and HAO, enzymes critical for NH_4^+ oxidation to $\text{NO}_2^-/\text{NO}_3^-$. Consequently, the nitrification process is slowed, leading to higher NH_4^+ concentrations in effluent. Additionally, high salt levels may favour the growth of halophilic bacteria (e.g., Halomonas), which lack efficient nitrifying capabilities, further reducing nitrogen removal efficiency. The competition for carbon sources under saline stress leads to synthesizing compatible solutes like glycine betaine, diverting carbon away from nitrogen removal functions, thus promoting ammonia accumulation (Vyrides and Stuckey, 2017). Interactions between salt, pH, DO, and other parameters, such as the C/N ratio and HRT, exacerbate the inefficiency of ammonium removal. Specifically, a higher C/N ratio can disrupt the balance between nitrifying and denitrifying bacteria. Reduced DO in high salinity can create hypoxic zones, inhibiting aerobic nitrification (Huang et al., 2024).

For TN_{out} , salinity inhibits the activity of key enzymes involved in denitrification, such as Nar and Nir activity. It alters the structure of cytochrome c in the electron transport chain, reducing electron transfer efficiency (Luo et al., 2021). As a result, $\text{NO}_3^-/\text{NO}_2^-$ reduction is hindered, and TN removal is impaired. Additionally, high salt concentrations increase the competition for carbon sources, creating an imbalance in the C/N ratio (Ruan et al., 2020). This issue disrupts the availability of electron donors for denitrification, leaving nitrogen oxides (NO_x^-) in the effluent. The accumulation of NO_2^- due to inhibited nitrification can feedback to suppress AOB activity, creating a vicious cycle of reduced nitrogen removal. Interactions with other parameters, such as DO, pH and temperature, further amplify these effects. Specifically, lower DO can hinder aerobic processes, while an optimal pH and temperature are crucial for maintaining effective nitrogen removal (Mishra et al., 2022).

3.6. Implications and outlook

This study used interpretable machine learning to predict nitrogen removal in MBR systems treating high-salinity wastewater. CatBoost model showed significant improvements in R^2 (9 %) and RMSE (14 %) for NH_4^+ - N_{out} , and similar gains for TN_{out} . SHAP analysis highlighted salinity as a key factor, disrupting nitrification and denitrification, while COD_{eff} , DO and C/N ratio influenced microbial activity. Prior studies have consistently demonstrated the inhibitory effect of high salinity on ammonia oxidation by ammonia-oxidizing bacteria (AOB) (Shi et al.,

2021; Xu et al., 2024). The SHAP analysis corroborates this but adds depth by revealing how salinity interacts with other key parameters, such as COD removal efficiency. Specifically, we found that high COD removal exacerbates carbon competition and starving nitrifiers of essential substrates, a nuanced finding not emphasized in prior work. Research on denitrification under saline conditions has shown that salinity impairs Nar activity (Li et al., 2022; Zhang et al., 2015). Salinity inhibits Nar and alters the microbial community structure, further impairing carbon transfer to anoxic zones. Previous studies have explored this interplay less, often focusing on single stressors.

The temperature-related findings also align with previous research but provide more operationally relevant insights. Luo et al. (2019) examined temperature effects on nitrification in non-saline systems but did not account for the complex interactions with salinity. The SHAP waterfall plots indicate that below 20 °C, Nar activity decays significantly due to Arrhenius-type behaviour, but this effect diminishes above 25 °C due to thermal inertia. This insight provides more specific guidance for operational strategies in saline wastewater treatment. Furthermore, while studies on aeration optimization have focused on improving DO for nitrogen removal (Zhang et al., 2019), this research highlights specific DO thresholds (0.5–1.2 mg/L) for optimal performance under salinity stress. This result contrasts with prior work that provided general aeration guidelines, offering a more precise operational range.

This study's model has some limitations for direct application in real-world environments. Extensive calibration is required when applying the model to wastewater with different compositions and for potential variations in plant-specific operational conditions (e.g., fluctuations in salinity levels). Continuous monitoring of environmental parameters is also necessary to ensure reliable model predictions. Additionally, the study focuses solely on high-salinity wastewater, and further adaptations may be needed for other types of influent. Future research could explore the combined effects of salinity with other stressors like heavy metals (Arévalo et al., 2013) or organic pollutants (Holloway et al., 2014), as these could compound the challenges of nitrogen removal. Factors such as adsorbents and membrane fouling could be incorporated into future models to refine operational strategies (Xu et al., 2024). Further validation under real-world conditions and incorporation of additional variables will strengthen model applicability and robustness.

4. Conclusion

This study leveraged interpretable machine learning to predict nitrogen removal in a MBR treating high-salinity wastewater. CatBoost achieved the best performance with R^2 of 0.88–0.91 and RMSE of 4.27–4.35 for NH_4^+ - N_{out} and TN_{out} . Compared to the second-best model, it improved R^2 by 9 %-10 % and reducing RMSE by 14 %-16 %, respectively. SHAP analysis identified salinity as the dominant factor, disrupting nitrification via enzyme inhibition and denitrification through carbon metabolic competition. COD_{eff} and DO governed microbial activity, while temperature modulated denitrifier enzyme kinetics. PDPs further revealed nonlinear thresholds in salinity and C/N ratio effects. These findings demonstrate how multi-faceted explainability tools can decode complex interactions in biological systems. Targeted operational adjustments, such as salinity control and carbon supplementation, can also be guided to optimize nitrogen removal. Interpretable machine learning bridges predictive modelling and mechanistic understanding, offering actionable strategies for enhancing MBR performance in high-salinity wastewater treatment.

CRedit authorship contribution statement

Qing Wei: Writing – original draft, Methodology. **Zuxin Xu:** Supervision, Funding acquisition. **Hailong Yin:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2025.132393>.

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